

BESS Project

Commissioning & SAT report

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1. Configuration

Table 1.1 Energy storage system configuration

Proposal	Container	Container quantity	PCS	Step-up transformer	Number of PCS-skid	EMS system	Capacity

2. Preparations

Table 2.1 Equipment inspection

No.	Check items	Requirement	Result
1	PCS skid	The appearance color is correct, with no noticeable color difference. The LOGO and markings are accurately in compliance with the drawing requirements, with no paint damage. The screen printing and labels show no damage, peeling, or bulging. The bolt surface has clear markings, and the internal equipment is installed correctly with neat wiring that is securely tied up. The interior is clean, free of residual materials and auxiliary items, and there is no significant dust. The paint protection is good, and the door seal strip is properly compressed, ensuring a good closure	
2	Battery Container	The appearance color is correct, with no noticeable color difference. The LOGO and markings are accurately in compliance with the drawing requirements, with no paint damage. The screen printing and labels are undamaged, without peeling or bulging. The bolt surface has clear markings. Internal equipment is installed correctly, with neat wiring that is securely tied up. The interior is clean, free from residual materials	

		and auxiliary items, and there is no significant dust. The paint protection is good, and the door rubber strips are properly compressed, ensuring a good seal	
3	Combiner Cabinet	The appearance color is correct, with no noticeable color difference. The LOGO and markings are accurately in compliance with the drawing requirements, with no paint damage. The screen printing and labels show no damage, peeling, or bulging. The bolt surface has clear markings. Internal equipment is installed correctly, with neat wiring that is securely tied up. The interior is clean, free from residual materials and auxiliary items, and there is no significant dust. The paint protection is good, and the door seal strip is properly compressed, ensuring a good closure	
4	Fire-fighting equipment	The appearance color is correct, with no noticeable color difference. The LOGO and markings comply with the drawing requirements, with no paint damage. The screen printing and labels show no damage, peeling, or bulging. The bolt surface has clear markings. Internal equipment is installed correctly, with neat wiring that is securely tied up. The interior is clean, with no residual materials or auxiliary items, and no obvious dust.	

		The paint protection is good. The door seal strip is properly compressed and closes well	
5	Liquid cooling unit	The appearance color is correct, with no noticeable color difference. The LOGO and markings are accurately in compliance with the drawing requirements, with no paint damage. The screen printing and labels are undamaged, without peeling or bulging. The bolt surface has clear markings. Internal equipment is installed correctly, with neat wiring that is securely tied up. The interior is clean, free from residual materials and auxiliary items, and there is no significant dust. The paint protection is good, and the door seal strip is properly compressed, ensuring a good closure	
6	High-voltage box	Check whether the surface of the high pressure box is damaged, paint off, and whether the mark is correct	
7	Battery cluster power line and communication power harness	Whether the wiring harness is damaged and whether the communication is normal	

Table 2.2 Electrical safety inspection

No.	Check items	Requirement	Result
1	The terminal L and N of the Combiner Cabinet are connected to the main power	Use the continuity function of the multimeter to test whether L and N of the junction box are short circuit	
2	UPS 220V AC Test	Test for short circuit at UPS220VAC	
3	Combiner Cabinet 12V DC	Test whether 12VDC is short circuit	
4	Combiner Cabinet 24V DC	Test whether 24VDC is short circuit	
5	The input voltage between B+ and B- in the high voltage box	The voltage between B+ and B-is consistent with the voltage between the total positive and total negative of the battery cluster;	
6	The output voltage in the high voltage box	There is no short circuit between positive and negative, and no reverse connection	
7	Insulation test	Use an insulation meter to measure the insulation resistance between the positive and negative terminals of the battery cluster and the shell or frame; 2Measure the resistance value of the battery cluster to ground and negative to ground is not less than 2MΩ.	

3. Commissioning

Table 3.1 Commissioning data sheet

No.	Item	Remarks
1	BMS requirement confirmation form	
2	Combiner cabinet drawings	
3	Network topology	
4	Cable inventory	
5	Fire communication topology	
6	Battery container user manual	
7	Liquid cooling specification	
8	Fire specification	
9	BMS specification	

Table 3.2 Tools

No.	Name	specifications	quantity	Remarks
1	notebook PC		as required	
2	CAN box		as required	
3	multimeter	DC1500V	as required	
4	insulation tester		as required	
5	DC clamp meter		as required	
6	infrared thermometer		as required	
7	Tool kits		as required	
8	electric screw driver		as required	

9	The equalizer		as required	
10	Liquid replenishment device		as required	
11	Insulating shoe		as required	
12	insulating gloves		as required	
13	USB-RS485 converter		as required	
14	European standard socket converter		as required	

3.1 Technical requirements

In order to ensure the smooth commissioning of the system, the installation needs to meet the following technical conditions:

1) Cold commissioning stage

- Temporary power supply at the site: 220V, frequency 50Hz, power>5kw
- Power cables from the battery side to the PCS DC side are connected .
- Communication cables are connected.
- Power supply(Fire/Liquid cooling system): 380V, frequency 50Hz, power>10kw.

2) Hot communication stage

- Grid power is available.

Table 3.3 Commissioning Schedule

No.	Content	Construction period (days)	personnel
1	Cold commissioning	Energy storage battery system cold commissioning	
		PCS cold commissioning	
2	Communication test	BMS and EMS communication test	
		PCS and EMS communication test	
		Transformer control box and EMS communication test	
		Static communication between EMS and grid	
3	Fire commissioning	Fire control software confirmation	
		On-site fire control system communication test	
		On-site fire control logic test	
		Central station fire control test	
4	The whole station communication test	EMS and BMS communication test	
		EMS and transformer communication test	
		EMS and PCS communication test	
		Communication between EMS and grid	
5	Hot commissioning	Charge and discharge	

Table 3.4 Personnel

Personnel sign-in sheet			
No.	name	telephone	remarks
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

3.2 Battery cold commissioning

Prior to the commencement of formal commissioning, the following verification and tightening procedures shall be performed:

- 1) Check and re-tighten all secondary wiring connections in the combiner cabinets and packs using a designated screwdriver.
- 2) Inspect and re-torque all electrical structure bolts within electrical cabinets and combiner cabinets using specialized tools.
- 3) Ensure that the container grounding is reliable and conforms to the required specifications.

3.2.1 General inspection

1) Visual Inspection

The container and cabinet exteriors shall be free from damage, with all equipment properly installed and no signs of fluid leakage.

2) Safety Signage

Prominent safety labels must be installed at all locations with potential hazards.

3) Battery Rack Assembly

All battery packs and the high-voltage box must be fully installed and secured. Every fixing bolt shall be individually checked for tightness and absence.

4) Combiner Cabinet Assembly

Internal equipment must be securely mounted. Circuit breaker arc chutes shall be completely installed, and busbar protective covers must be correctly positioned to prevent direct human contact with power cables and busbars.

5) Cable Entry Seal Inspection

Fireproof sealing for cable entry holes at the container base must be intact and in good condition.

6) Interior Lighting Inspection

Lighting fixtures, emergency lights, light switches, and sockets shall be properly installed and functioning correctly.

7) Firefighting Equipment Inspection

The fire alarm control panel, temperature/smoke detectors must be securely installed and present in the required quantities. Portable fire extinguishers must be complete in number and show normal pressure readings.

8) Insulating Mat Installation

Insulating mats must be laid along main walkways inside the container, with no damage present.

9) Door Inspection

All container and cabinet doors must open and close smoothly, with functional locks.

3.2.2 Construction wiring inspection

1) Power and Communication Cables

Cable specifications and models comply with design requirements. Wiring phase sequence and polarity are correct, connections are tight, and installation is performed to standard.

2) Cable Labeling Inspection

Cables are correctly and clearly labeled at both ends.

3) Container Grounding Inspection

The container grounding point is securely connected to the ground grid.

4) Main Circuit Wiring Inspection between Battery PACKs

All connections are present, wiring matches the diagram, and crimps are secure.

5) High-Voltage Box Interface Inspection (Power Terminals)

P+/P- and B+/B- terminals are correctly connected and tightly secured.

6) High-Voltage Box Interface Inspection (Auxiliary Terminals)

AC 220V power, DC 24V power, and CAN communication connectors are properly engaged and show no looseness.

7) Combiner Cabinet Wiring Inspection

Input and output cables for the busbars have correct polarity. Bolts are tightened to specification, and secondary wiring is correct and secure.

8) Control Cabinet Wiring Inspection

All internal cables are correctly connected, and all bolts are properly torqued.

3.2.3 Battery system inspection

1) Battery cluster voltage test

Requirement:

Before initial test, ensure the incoming power supply is disconnected, the lower branch unit is closed, and the UPS is powered off. Verify that there are no loose connections or short circuits in the overall cabinet circuit, and confirm that grounding is reliable.

Procedure:

Measure the terminal voltage of each battery string to verify it is within the normal range and check that power cable connections are secure.

Record the battery string voltage value (a) displayed on the BMS screen.

Use a multimeter to measure the actual battery string voltage (b).

Compare the difference between values (a) and (b).

2) Battery cluster insulation test

Procedure:

Use an insulation tester to measure the insulation resistance between the positive and negative poles of the battery string and the casing or battery rack.

Verify that the resistance values between the positive pole to ground and negative pole to ground are not less than 2 MΩ.

3) Battery cluster contactor test

Procedure:

Power on the system and start the display screen.

Observe whether the voltage of each battery string is within the normal range via the display.

Check the BMS alarm interface to confirm that no alarms are present.

Use the one-touch grid-connection start function on the display screen.

Observe whether the merging terminal of each string closes properly.

Use the one-touch grid-connection shutdown function on the display screen.

Observe whether the main merging segment of each string opens properly.

4) Voltage sampling test

Procedure:

Power on the system and start the display screen.

Check the total stack voltage and individual cell voltage data displayed on the screen.

Verify that the acquired total stack voltage, average voltage, and cell voltages are accurate and correctly displayed.

Record the average voltage, maximum cell voltage, and minimum cell voltage in a table.

If the voltage deviation exceeds 0.05 V, record the location of the cell with the lowest voltage.

5) Temperature sampling test

Procedure:

Power on the system and start the display screen.

Check the temperature data of each individual cell on the display.

Confirm that the acquired temperatures—including the maximum and minimum cell temperatures—are accurate and correctly displayed.

Record the average temperature, maximum cell temperature, and minimum cell temperature in a table.

6) Auxiliary Equipment Check

Procedure:

Verify the data from the temperature and humidity sensor.

Confirm the data from the dehumidifier.

Check the fire protection system data.

Verify the liquid cooling system data.

7) Software Version Check

Procedure:

Check the version numbers of the main and slave controllers on the display screen.

Confirm that the on-site program matches the project program.

3.3 PCS cold commissioning scheme

Table 3.5 PCS inspection content and duration

No.	Item		Construction period (days)
1	PCS preliminary inspection	Inspect appearance and components	
		Check and re-tighten the electrical and structural bolts	
		Check the secondary circuit wiring is correct and reliable	
2	Secondary circuit commissioning	Confirm the program version	
		Communication and display confirmation	
		Confirm settings, parameter and pressure plate	
3	PCS and BMS integrated commissioning	Check the DC voltage sampling	
		Check the communication between PCS and BMS	
		Confirm the working status of each branch of PCS	
4	Communication Cabinet / Low-Voltage Cabinet Commissioning	Confirm the power supply for the switch box	
		Confirm the power supply for the secondary circuit of the high voltage cabinet	
		Ground knife/circuit breaker manual opening and closing test	
		Remote/local electric separation test	

1) Prior to PCS functional test:

Verify that the DC cable connections between the battery container and the PCS, as well as the dry contact signal cables and communication cables, have been completely installed.

The manufacturer shall assist the construction team in verifying the correct polarity of the DC cable connections (positive and negative).

The manufacturer shall support the construction team in inspecting whether the connection between the PCS equipment and the main grounding grid is compliant and reliable.

Confirm that the upper equipment on the high-voltage side is not energized.

2) PCS Preliminary Inspection

Inspect the cabinet exterior to confirm there is no deformation, damage, or insecure fastening.

Inspect all major internal components of the cabinet to identify any damage or loose connections caused by transportation or on-site installation.

Check that all electrical and structural bolts at the PCS busbars and grounding points are tight; re-tighten them using a designated torque tool.

Verify the reliability of wiring in the PCS secondary circuits and the low-voltage side secondary circuits of the high-voltage cabinet. With the equipment de-energized, re-tighten all secondary circuit connections.

3) Secondary circuit commissioning

Use a temporary power supply to energize the secondary circuits of the DC compartment combiner cabinet, delivering battery compartment DC voltage to the PCS DC side. The PCS will power up using this DC supply. Connect a laptop with the host computer software to the PCS controller to confirm the software version and equipment operating status.

Test and verify that all display and control functions on the LCD interface operate correctly, and confirm that the indicators on the front panel display properly. Check that the acquisition, display, and output control of relevant monitoring signals are correct.

Use the host computer software to set operating values, parameters, and control

function settings (plate settings), and confirm that all parameters are correctly configured.

4) PCS and BMS integrated commissioning

Use a temporary power supply to energize the secondary circuits of the DC compartment combiner cabinet, delivering battery compartment DC voltage to the PCS DC side. The PCS powers up using this DC supply.

Use the PCS touchscreen to set the communication parameters between the PCS and BMS.

Use the PCS touchscreen to check the communication status between the PCS and BMS.

Under static conditions, verify via communication that the total voltage, SOC, and SOH data for each branch are consistent.

5) Communication Cabinet / Low-Voltage Cabinet Commissioning

Power up the communication cabinet via its UPS.

Confirm that equipment such as switches in the communication cabinet are powered normally and that equipment indicators show no abnormalities.

Confirm that the power supply for the secondary circuits on the low-voltage side of the high-voltage room is normal, with no abnormal indicator displays.

Manually operate the grounding switch and circuit breaker using the manual handle to verify normal operation. Use the host computer software of the transformer station's monitoring system to confirm correct position change indication for the grounding switch and circuit breaker.

Set the grounding switch to the open position, close all high-voltage room cabinet doors, and confirm there are no electrical interlock faults.

Use the local buttons to open and close the circuit breaker, confirming that the electrical open/close functions operate normally.

Use the EMS system to remotely control the circuit breaker, confirming that the remote open/close functions operate normally.

3.4 Static communication test scheme

Table 3.6 Static communication content and duration

No.	Item	Construction period (days)
1	The static communication between BMS and EMS	
2	The static communication between PCS and EMS	
3	The static communication between the transformer and EMS	
4	The static communication between EMS and grid	

1) Precautions

Before conducting static communication tests, power up the communication equipment in the communication cabinet using the UPS in the PCS power conversion system communication cabinet. Power up the BMS equipment in the battery compartment using the UPS in the combiner cabinet (for initial UPS startup, temporary AC power is required).

2) Static communication test between BMS and EMS

The BMS supplier shall load the IEC 61850 configuration file with the project-specified ID names onto the main controller.

The EMS supplier shall import the ICD file provided by the BMS supplier.

Use the EMS to view BMS static data, including string voltages, temperatures, SOC, etc.

Verify that the static data displayed by the EMS matches the actual collected data.

3) Static communication test between PCS and EMS

The PCS supplier shall load the IEC 61850 configuration file with the project-specified ID names onto the communication mainboard.

The EMS supplier shall import the ICD file provided by the PCS supplier.

Use the EMS to view PCS static data, including total string voltages, temperatures, SOC, etc. The PCS supplier shall verify that the static data displayed by the EMS matches the actual collected data.

4) Static communication test between EMS and transformer

The EMS shall configure the database according to the gateway protocol.

Use the EMS to view static data of the PCS power conversion system, including circuit breaker positions, grounding switch data, etc.

5) Static communication test between EMS and grid

The EMS shall configure the 104 forwarding channel and set the IP address as required by the system supplier.

Simulate and verify communication protocol data.

3.5 Fire control system cold commissioning

Table 3.7 Static communication content and duration

No.	Item	Construction period (days)
1	Confirm the on-site fire control host software version	
2	Confirm the status of local fire control host equipment	
3	On-site fire control unit control logic test	
4	Central station fire control host control logic test	

1) Precautions: Before conducting on-site fire protection tests, strictly perform the following sequence: de-energize the system, disconnect the relays and activation circuit wiring, then re-energize to perform the tests.

After completing the fire tests and confirming all fire system functions are normal, reconnect the activation circuit wiring and relays to restore the fire protection system to operational status.

2) Software Verification

Check the software version via the fire panel settings or using the host computer software.

Compare it with the engineering application software version.

3) Fire panel sensor status verification

Power on the fire alarm control panel.

Check the operational status of the aspirating smoke detectors locally on the fire panel.

Check the operational status of the heat detectors locally on the fire panel.

Check the operational status of the smoke detectors locally on the fire panel.

Check the operational status of the explosive/combustible gas detectors locally on the fire panel.

Check the operational status of the CO gas detectors locally on the fire panel.

Verify that other auxiliary devices on the fire panel are operating normally with no abnormal alarms.

4) Local fire panel control logic test

Requirement:

Power on the fire system, **ensuring it will not initiate fire suppressant agent discharge.**

This can be achieved by, for example, installing a short-circuit protection wire across the activation circuit terminals connecting the controller to the aerosol fire suppression units. Simultaneously, disconnect the actuator from the valve release mechanism so that even if the controller sends an activation signal, it will not physically trigger aerosol release. After debugging is complete, verify no activation signal is present, then remove the short-circuit protection wire, correctly reconnect the activation circuit to the controller, and restore the connection between the actuator and the valve mechanism.

Steps:

Individually trigger a smoke detector alarm and a heat detector alarm separately, causing the fire panel to indicate a single fire alarm condition.

Observe whether the actions of the audible/visual alarms, the central fire panel, and the BMS system meet the set requirements.

Simultaneously trigger both a smoke detector and a heat detector alarm, causing the fire panel to indicate a combined fire alarm condition, and wait for the 30-second delay period to complete.

Observe whether the actions of the audible/visual alarms, the central fire panel, and the BMS system meet the set requirements.

Press the manual call point button, check if the alarm sounders activate, and check if the fire panel alarms normally.

5) Ventilation System Controller Test

Check the display screen to verify the combustible gas and carbon monoxide display interfaces are functioning correctly.

Switch the manual/auto mode selector to manual, and check if the louvers open.

Set the manual start/stop selector to manual, and check if the fan starts.

6) Central fire panel Logic Test

This item depends on whether the fire protection of the whole station is supplied by our company.

3.6 Dynamic communication scheme

Table 3.8 Dynamic communication content and duration

No.	Item		Construction period (days)
1	BMS and EMS dynamic communication	Check dynamic battery data	
		Check dynamic liquid cooling system data	
		Check dynamic fire system data	
		Check dynamic temperature and humidity data	
		Check dynamic air conditioning data	
		Check BMS remote control and grid connection function	
2	PCS and EMS dynamic communication test	Check dynamic telemetry data	
		Remote control function test	
3	Transformer and EMS dynamic communication	Check high voltage side circuit breaker data	
		Check transformer thermostat data	
4	EMS and grid dynamic communication	Check dynamic telemetry data	
		Check dynamic telemetry data	

Note: After initial site energization, when the battery container, PCS power conversion system, and EMS are powered on, first verify dynamic communication data, then test dynamic control functions. BMS and EMS dynamic communication function.

1) BMS and EMS dynamic communication

Verify that the values and quantities for each battery string's total voltage, SOC, SOH, cell voltages, and cell temperatures match the on-site readings.

Verify that the operational status (on/off), operating mode, ambient temperature, outlet/return water temperature, and outlet/return water pressure of the battery compartment liquid cooling system match the on-site dynamic values.

Verify fire detection communication functionality is normal and that all fire protection sensors are operating correctly.

Verify that the temperature and humidity data from each collection point match the actual on-site conditions.

Verify that the HVAC coil temperature, humidity, room temperature, condensation temperature, and operational status data match the actual conditions.

After confirming local grid-connection operation via the BMS display is normal:

Use the EMS to remotely execute the one-touch start/grid-connection command for each branch, testing if the function operates correctly.

Use the EMS to remotely execute the one-touch stop/grid-disconnection command for each branch, testing if the function operates correctly.

2) PCS and EMS dynamic communication

Verify dynamic PCS telemetry data, including ambient temperature, insulation resistance, DC voltage, AC voltage, and IGBT temperatures.

Verify that the position change signals for the PCS DC circuit breaker and AC circuit breaker are normal.

Use the EMS to remotely initiate the start function for each branch, confirming the PCS remote control functionality for each branch is normal.

Set the PCS to remote mode.

Ensure all battery compartment branches are successfully connected to the grid.

Test the charge/discharge functions locally via the PCS touchscreen to confirm normal operation.

Use the EMS to issue charge/discharge commands to each branch, verifying that the issued commands correspond to the actual actions performed.

3) Transformer and EMS dynamic communication

Verify that the position data for the high-voltage side grounding switch, circuit breaker, and local/remote selector switch match the on-site status.

Verify that the dynamic data for the transformer core temperature matches the real-time data collected by the local temperature controller.

4) EMS and grid communication

Based on the point list, verify data in real-time with the dispatch system, ensuring values and units are consistent and channel transmission is stable.

3.7 Hot commissioning

Table 3.9 Hot commissioning scheme

No.	Item		Construction period (days)
1	Single container charge and discharge test	Charge and discharge test	
		Data collection	
		Record analysis	
2	whole station Charge and discharge test	Full power charge and discharge test	
		Data collection	
		Record analysis	
		System efficiency calculation	

After site energization, completion of dynamic communication tests, and fire protection system commissioning, proceed with system hot commissioning. Single-container charge/discharge test verifies EMS remote control and setpoint functions for each branch, and validates BMS charge/discharge logic and liquid cooling system performance. Whole-plant charge/discharge test primarily evaluates EMS whole-plant power control functionality, records total plant charge/discharge energy, and calculates conversion efficiency and system efficiency.

Note: During single-container charge/discharge, monitor the liquid cooling system outlet water pressure. Adjust via fluid filling/draining operations to maintain operating outlet pressure within the reference range.

1) Single-Container Charge/Discharge Test

Test Steps:

Check in the EMS system if communication with the liquid cooling system is normal, its operational status is normal, and there are no abnormal alarms.

Check in the EMS system if communication with the fire protection system is normal, its operational status is normal, and there are no device alarms.

Check in the EMS system if communication with the temperature and humidity sensors is normal, and verify the actual values are within the normal range.

Check in the EMS system if communication with the water leak sensors is normal and there are no abnormal alarms.

Gradually perform ramp-up power charging steps at 0.1P, 0.25P, and 0.33P. Maintain each power level for 5 minutes, observe if equipment status is normal, and record real-time values.

Gradually perform ramp-up power discharging steps at 0.1P, 0.25P, and 0.33P. Maintain each power level for 5 minutes, observe if equipment status is normal, and record real-time values.

After half an hour of charging at rated power, record the charging current and temperature difference data.

After half an hour of discharging at rated power, record the discharging current and temperature difference data.

Record the actual operating outlet/return water pressure and the reference pressure range for the liquid cooling system.

Record the maximum cell temperature, minimum cell temperature, and temperature differential range during charging, discharging, and idle states. (General reference values: Cell temperature 20°C - 40°C, Temperature differential $\leq 5^{\circ}\text{C}$).

Record the IGBT temperature range during charging, discharging, and idle states (within the technical agreement's ambient temperature requirements, not exceeding the PCS IGBT derating protection threshold).

Record the transformer winding temperature and core temperature operating range during charging, discharging, and idle states (not exceeding the alarm thresholds set by the transformer temperature controller).

Records:

After completing the stepwise ramp-up charging commands, observe the battery system. When the maximum cell voltage reaches the Level 3 alarm threshold, verify if the High Cell Voltage Level 3 alarm is triggered according to the strategy and associated power reduction occurs. Verify if the High Cell Voltage Level 2 alarm is triggered and associated standby operation occurs. Record the actual strategy execution and cross-check consistency with the BMS Customer Requirements Specification.

After completing the stepwise ramp-up discharging commands, observe the battery system. When the minimum cell voltage reaches the Level 3 alarm threshold, verify if the Low Cell Voltage Level 3 alarm is triggered according to the strategy and associated power reduction occurs. Verify if the Low Cell Voltage Level 2 alarm is triggered and associated standby operation occurs. Record the actual strategy execution and cross-check consistency with the BMS Customer Requirements Specification.

After charge/discharge cycles, statistically record the charge/discharge energy readings from both the BMS and PCS sides.

After discharge ends and the system rests for half an hour, record the maximum cell voltage, minimum cell voltage, average voltage, voltage differential, and note the location of the cell with the lowest voltage.

2) Whole-station Charge/Discharge Test

Test Steps:

Start with the entire plant system in a fully discharged state (based on cell voltage values).

For the whole plant, gradually execute ramp-up power charging steps at 0.1P, 0.25P, and 0.33P. Maintain each power level for 5 minutes. Continue charging at 0.33P until fully charged, recording the actual total plant power value.

For the whole plant, gradually execute ramp-down power discharging steps at 0.1P, 0.25P, and 0.33P. Maintain each power level for 5 minutes. Continue discharging at 0.33P until fully discharged, recording the actual total plant power value.

Records:

Record all running records during the whole-plant charge/discharge cycle.

Calculate the DC-side conversion efficiency based on the BMS-side energy statistics.

Calculate the PCS conversion efficiency based on the PCS-side energy statistics.

Calculate the overall system efficiency based on the grid interface meter energy statistics.

4. SAT Reports

4.1 Battery container Energization Inspection

Project name			
Item	Battery container Energization Inspection Record Form		
Battery container number		Personnel/date	
Scope	Requirements		Conclusion
Electrical inspection	Tighten and re-tighten electrical structural bolts		
Wire inspection	Re-tighten the secondary wiring of the combiner cabinet and the pack to check if they are reliable		
Appearance inspection	Check that the appearance of the container and the cabinet body is undamaged, the equipment is installed normally, and there is no leakage.		
Safety signs	All locations with potential safety hazards should be marked with prominent safety signs.		
Battery rack assembly	Check whether the battery PACK and high-voltage box are all installed in place, and whether the fixing bolts are tightened without any missing parts.		
Assembly of the manifold cabinet	Check whether the equipment inside the cabinet is installed firmly, whether the arc isolation plates of the circuit breakers are fully installed, whether the copper bar protection plates are installed in place, and avoid direct human contact with power cables and copper bars.		

Check the cable seal	The fireproof sealing of the inlet hole at the bottom of the box is in good condition.	
Check the lighting inside the box	Lighting fixtures, lighting switches, sockets and so on are installed in place and function normally.	
Fire equipment inspection	The fire control host and smoke detection detector are installed firmly and in full quantity. The number of hand-held fire extinguishers is complete and the pressure is normal.	
Check the door	Check the door is smooth and the door lock is normal.	
Construction wiring inspection		
Cable inspection	The specifications and models of power cables and communication cables meet the requirements of design drawings, the wiring sequence and polarity are correct, the cable connection is tight, and the construction wiring is standard.	
Check the label	The cable is marked at both ends, and the marking is clear and correct.	
Check the grounding of the box	The container connection point is reliably connected to the ground network.	
Check the main circuit wiring	There is no missing wiring, the wiring mode is consistent with the wiring diagram, and the crimping is reliable.	
Check the high voltage box interface	P+/-and B+/-wiring is correct and securely in place.	

Check the high voltage box interface	High voltage box AC220V power supply interface, DC24V power supply interface, CAN communication interface and other connectors are in place without loosening.	
Check the wiring of the junction box	The polarity of the input and output cables of the copper busbar is correct, the bolts are tightened, and the secondary wiring is correct and not loose.	
Check the wiring of the control cabinet	All internal cables are correctly wired and bolts are tightened.	
Signing page		

4.2 Battery cold commissioning table

Project name							
Item	Battery cold commissioning table						
Battery container number		Personnel/date					
Test items	test data						Test conclusions
Battery cluster voltage test	cluster 1		cluster 2		cluster 3		
	cluster 4		cluster 5		cluster 6		
	cluster 7		cluster 8		cluster 9		
Battery cluster insulation test	cluster 1		cluster 2		cluster 3		
	cluster 4		cluster 5		cluster 6		
	cluster 7		cluster 8		cluster 9		
Contactor control test	cluster 1		cluster 2		cluster 3		
	cluster 4		cluster 5		cluster 6		
	cluster 7		cluster 8		cluster 9		
Voltage acquisition function test	Maximum voltage of a single unit						

	Minimum voltage of a single unit		
	average voltage		
Temperature collection function test	maximum temperature		
	minimum temperature		
	average temperature		
software version			
Signing page			

4.3 PCS cold commissioning table

project name			
Item	PCS cold commissioning table		
PCS number		Personnel/date	
Test items			Results Test conclusions
PCS preliminary inspection	Inspect appearance and components		
	Check and re-tighten the electrical and structural bolts		
	Check the secondary circuit wiring is correct and reliable		
Secondary circuit commissioning	Confirm the program version		
	Communication and display confirmation		
	Confirm settings, parameter and pressure plate		
PCS and BMS integrated commissioning	Check the DC voltage sampling		
	Check the communication between PCS and BMS		
	Confirm the working status of each branch of PCS		
Communication Cabinet /	Confirm the power supply for the switch box		
Low-Voltage Cabinet	Confirm the power supply for the secondary circuit of the		

Commissioning	high voltage cabinet	
	Ground knife/circuit breaker manual opening and closing test	
	Remote/local electric separation test	
Signing page		

4.4 Fire control system cold commissioning table

project name			
Item	Fire control system commissioning record form		
Battery container number		Personnel/date	
Test items	Test data		Test conclusions
On-site fire control host software version confirmation			
Confirm the status of local fire control host equipment	A suction probe		
	hygrosensor		
	Smoke detector		
	Combustible gas detection		
	CO gas detector		
	accessory equipment		
On-site fire control unit control logic test	single fault		
	Composite failure		
	Manual Fire Alarm Button		
	Exhaust system		
Logic test of the main station fire control host	software release		
	Communication function		
	Alarm function		

	history		
Signing page			

4.5 Communication test

project name			
Item	Communication test record sheet		
Equipment number		Personnel/date	
Content	Results		Test conclusions
The static communication between BMS and EMS is normal			
The static communication between PCS and EMS is normal			
The static communication between the transformer and EMS			
The static communication between EMS and scheduling system is normal			
BMS and EMS dynamic communication function	Check dynamic battery data		
	Check dynamic liquid cooling data		
	Check dynamic fire data		
	Check dynamic temperature and humidity data		
	Check dynamic air conditioning data		
	Check BMS remote control and grid connection function		
PCS and EMS dynamic communication	Check dynamic telemetry data		
	Check dynamic telemetry data		
	Remote control function test		

function	Dynamic remote adjustment instruction test		
The static communication between the transformer and EMS	High voltage side circuit breaker data		
	Transformer thermostat data		
EMS and grid	Simulate and verify communication protocol data		
Signing page			

4.6 Single container charge and discharge test

project name				
Item	Single container charge and discharge test			
Battery container number		Date of examination		
Data				Test conclusions
Communications				
Liquid cooling system		fire extinguisher system		
Temperature and humidity collection		Water immersion collection		
Charge and discharge test				
0.1P charge		0.1P discharge		
0.25P charge		0.25P discharge		
0.5P charge		0.5P discharge		
0.5P charge		0.5P discharge		
1P charge		1P discharge		
Current, temperature difference and liquid cooling pressure under maximum power should be recorded				
charge current		Charge temperature difference		
Discharge current		Discharge temperature difference		
Water outlet pressure		Refer to the pressure range		

Liquid cooling operating temperature range	Reference range 22°C-35°C			
IGBT operating temperature range				
Temperature range of transformer operation				
Charge and discharge logic records				
Capacity and cell information				
BMS/PCS charge capacity		BMS/PCS discharge capacity		
Minimum voltage		Maximum voltage		
Average voltage		Minimum voltage position		
Signing page				

4.7 Whole station Charge and discharge test

project name				
Item		whole station Charge and discharge test		
Battery container number		date of examination		
Data	Test conclusions			
Charge and discharge test of the whole station				
0.1P charge		0.1P discharge		
0.25P charge		0.25P discharge		
0.5P charge		0.5P discharge		
0.5P charge		0.5P discharge		
1P charge		1P discharge		
Time and SOC interval records				
Start time of charging		End time of charging		
Initial SOC of charging		End SOC of charging		
Start time of discharging		End time of discharging		
Initial SOC of discharge		End SOC of discharging		
Capacity				
BMS charge capacity		BMS discharge capacity		
PCS charge capacity		PCS discharge capacity		

System efficiency calculation	
BMS side efficiency	
PCS side efficiency	
Signing page	